1. Creating and Renaming Files/Directories

1. Create a directory named test_dir using mkdir.
2. Inside test_dir, create an empty file called example.txt.
3. Rename example.txt to renamed_example.txt using mv.

```
tanwi@LAPTOP-TM2GAE9R:~$ mkdir Linux_Basics_Tanwistha
tanwi@LAPTOP-TM2GAE9R:~$ cd Linux_Basics_Tanwistha
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ mkdir test_dir
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ touch test_dir/example.txt
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ mv test_dir/example.txt test_dir/renamed_example.txt
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ ls -l test_dir
total 0
-rw-r--r-- 1 tanwi tanwi 0 Oct 26 07:32 renamed_example.txt
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ ls
test_dir
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ ls -l
total 4
drwxr-xr-x 2 tanwi tanwi 4096 Oct 26 07:32 test_dir
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$
```

2. Viewing File Contents

4. Use cat to display the contents of /etc/passwd.

```
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
tanwi:x:1000:1000::/home/tanwi:/bin/bash
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$
```

5. Display only the first 5 lines of /etc/passwd using head.

```
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$
```

6. Display only the last 5 lines of /etc/passwd using tail.

```
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ tail -n 5 /etc/passwd
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
tanwi:x:1000:1000::/home/tanwi:/bin/bash
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$
```

3. Searching for Patterns

7. Use grep to find all lines containing the word “root” in /etc/passwd.

```
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$
```

4. Zipping and Unzipping

8. Compress the test_dir directory into a file named test_dir.zip using zip.

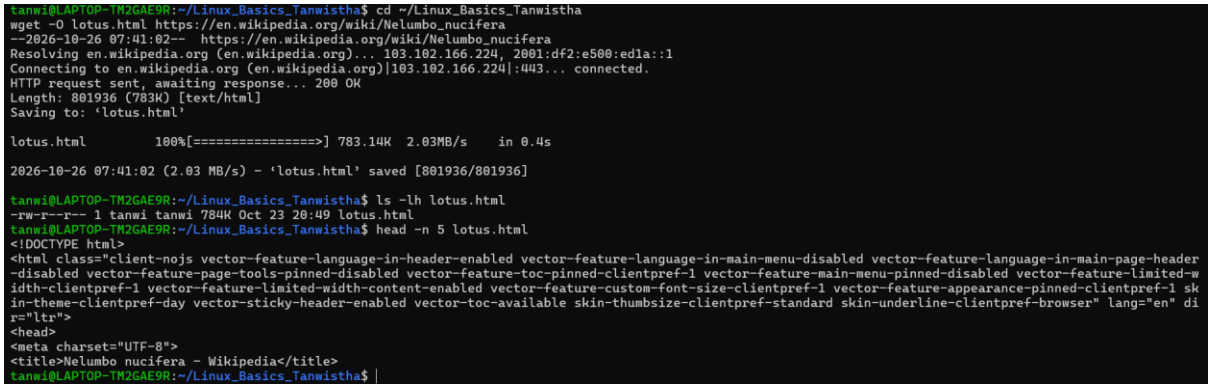
9. Unzip test_dir.zip into a new directory named unzipped_dir.

```
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ cd ~/Linux_Basics_Tanwistha
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ ls -l test_dir.zip
-rw-r--r-- 1 tanwi tanwi 352 Oct 26 07:37 test_dir.zip
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ mkdir -p unzipped_dir
unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
   creating: unzipped_dir/test_dir/
 extracting: unzipped_dir/test_dir/renamed_example.txt
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$ ls -R unzipped_dir
unzipped_dir:
test_dir

unzipped_dir/test_dir:
renamed_example.txt
tanwi@LAPTOP-TM2GAE9R:~/Linux_Basics_Tanwistha$
```

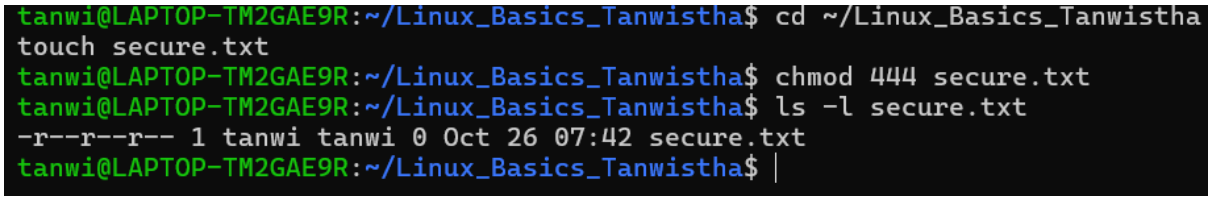
5. Downloading Files

10. Use wget to download a file from a URL (e.g., https://example.com/sample.txt).



6. Changing Permissions

11. Create a file named secure.txt and change its permissions to read-only for everyone using chmod.



7. Working with Environment Variables

12. Use export to set a new environment variable called MY_VAR with the value “Hello, Linux!”.

